

GreenStep Employee Sustainability Challenge

QA Activities Document

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1 Test Plan

1.1 Scope

This plan covers functional testing of the GreenStep Employee web application: authentication and route protection, onboarding, action logging (including note/photo bonus points and the duplicate-submission guard), streaks, the team-request and admin-approval workflow, leaderboards (team and individual, with tie handling and category/time filters), administrative tools (challenge creation, join-by-code, CSV export, admin-only access), and the error/404 pages. Performance, load, and security penetration testing are out of scope for this phase.

1.2 Test Environment

- **Application:** Next.js 15 build, run both locally (`npm run dev` at `localhost:3000`) and on the deployed Vercel instance.
- **Browsers:** Desktop Chrome and mobile-width viewport (the UI is mobile-first).
- **Auth:** Clerk test instance; an admin account with `publicMetadata.role = "admin"` and one or more standard accounts.
- **Database:** Neon PostgreSQL (development branch), inspected with Prisma Studio (`npm run db:studio`); seed scripts under `scripts/` provide demo data.

1.3 Approach and Responsibilities

Testing was **manual**, exploratory plus the scripted cases in Table 1, executed against two or more accounts to exercise multi-user flows (teams, leaderboards). Hadi led test execution and the final end-to-end pass; each team member tested the features they implemented, and defects were recorded as GitHub issues and fixed on feature branches. Pass/Fail reflects the current build; fixed defects are listed in Table 2 with their commit references.

2 Test Cases

Table 1: Functional test cases for the core flows.

ID	Feature	Preconditions	Steps	Expected	Actual	P/F
TC-01	Sign-up / sign-in	No session	Visit <code>/sign-up</code> , create account; sign out; sign in	Account created; redirected to <code>/dashboard</code>	As expected	Pass
TC-02	Route protection	No session	Request <code>/dashboard</code> directly	Middleware redirects to <code>/sign-in</code>	Redirected	Pass
TC-03	Onboarding	New user, first login	Complete onboarding (department)	<code>onboarded=true</code> ; lands on dashboard	As expected	Pass
TC-04	Log action	Signed in	Log “Biked to work”	Points added; appears in recent activity	As expected	Pass
TC-05	Note/photo bonus	Signed in	Log an action with a note and a photo	Base points +10 (note) +15 (photo)	Bonus applied	Pass
TC-06	Duplicate guard	Signed in	Submit the same action twice within 10s	First logs; second returns 409; only one record	One record; 409 on retry	Pass

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Table 1 continued

ID	Feature	Preconditions	Steps	Expected	Actual	P/F
TC-07	Streak increment	Logged yesterday	Log an action today	Streak = previous + 1	Incremented	Pass
TC-08	Streak gap-reset	Last action ≥ 2 days ago (seeded)	Log an action today	Streak resets to 1	Reset to 1	Pass
TC-09	Team request	In a challenge, no team	Submit a team request	Request saved PENDING ; shown to admin	As expected	Pass
TC-10	Approve request	Admin; a PENDING request	Approve it	Team + TeamMember created; status APPROVED	As expected	Pass
TC-11	Reject request	Admin; a PENDING request	Reject it	Status REJECTED ; no team created	As expected	Pass
TC-12	Leaderboard (team)	≥ 2 teams, some tied	Open Leaderboard, Team tab	Teams ranked desc; tied teams share a rank	Ties grouped	Pass
TC-13	Leaderboard (individual)	Several users with actions	Switch all-time / rising / streak / diverse; apply category + time filters	Each view ranks correctly; filters scope results	As expected	Pass
TC-14	Create challenge	Admin	Create a challenge	Saved with a unique 6-char code; creator auto-joined	As expected	Pass
TC-15	Join by code	Standard user	Enter a valid code; then an invalid code	Valid joins; invalid/expired rejected with message	As expected	Pass
TC-16	CSV export	Admin	Export participation CSV	CSV downloads with participation rows	File downloaded	Pass
TC-17	Admin-only access	Standard (non-admin) user	Attempt an admin action (create challenge)	Blocked — Forbidden (server action throws)	Blocked	Pass
TC-18	Error / 404	—	Visit an unknown route; trigger a server error	not-found / error page renders gracefully	As expected	Pass

3 Defect Reports

Table 2 lists defects found during development. Fixed defects cite the commit that resolved them; the final entries are known limitations tracked as open defects.

Table 2: Defect reports (fixed and open).

ID	Description	Severity	Steps to reproduce	Status
D-01	Clerk users synced twice, creating duplicate User rows	Med	Sign in repeatedly; observe duplicate users in DB	Fixed (b8d0f1a)
D-02	Dashboard crash from Prisma client / schema mismatch	High	Load <code>/dashboard</code> after a schema change without regenerating the client	Fixed (3ce2d8a, 8b7be7a)

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Table 2 continued

ID	Description	Severity	Steps to reproduce	Status
D-03	Leaderboard crash when a user/team had no related record (null)	High	Open leaderboard with a user not on a team	Fixed (87caf66)
D-04	Onboarding action strings did not match the catalog	Med	Complete onboarding; selected actions mismatched	Fixed (87caf66)
D-05	Clerk authentication redirect loop on dashboard	High	Sign in with mismatched Clerk env keys	Fixed (74ae5c9)
D-06	Duplicate action logs from rapid repeat clicks	Med	Tap “log” multiple times quickly	Fixed (1a48b02; 10 s serializable guard)
D-07	Tied leaderboard scores not sharing a rank	Low	Two teams/users with equal points ranked sequentially	Fixed (5db57ba; dense rank)
D-08	“All Activity” dashboard totals computed incorrectly	Med	Switch to All-Activity view; totals wrong	Fixed (f21e3fe)
D-09	Duplicate “GreenStep” option in the challenge selector	Low	Open challenge dropdown; global listed twice	Fixed (176e43b)
D-10	Challenge isolation / scoped-export logical gaps	Med	View badges/rank/exports across challenges	Fixed (28c382c)
D-11	No edit or undo of a logged action	Med	Log an action; attempt to edit/remove it	Open (planned)
D-12	Actions are self-reported (honor system, no verification)	Low	Log any action without proof	Open (by design)
D-13	Large <code>DashboardClient.tsx</code> component (~2,080 lines, reduced from ~2,300)	Low	Review the dashboard component size	Open (refactor planned: per-tab components)

4 Known Limitations

Defects D-11–D-13 above are the project’s known limitations, recorded as open defects for transparency: there is no edit/undo of a logged action; action data is self-reported on an honor-system basis with no verification; and the main dashboard component remains large and is slated for decomposition into per-tab components. These are also reflected in the To-Do / Future Enhancements section of the Project Report.

References

- [1] ISTQB, “Standard Glossary of Terms used in Software Testing,” v3.2, International Software Testing Qualifications Board, 2023.
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